

Normal Numbers Have Probability One: A Borel–Cantelli Proof

Let $b \geq 2$. Every x in $[0,1]$ is represented by its unique non-terminating base- b expansion $x = \sum X_n b^{-n}$, where the digits X_n are i.i.d. uniform on $\{0, \dots, b-1\}$ under Lebesgue measure.

Step 1: Frequency of a fixed digit.

Fix a digit a and define $I_n = 1\{X_n = a\}$. Then $E[I_n] = 1/b$ and $\text{Var}(I_n) = (1/b)(1-1/b)$. Writing $S_N = \sum_{n \leq N} I_n$, we have $E[S_N] = N/b$ and $\text{Var}(S_N) = N(1/b)(1-1/b)$. By Chebyshev, $P(|S_N/N - 1/b| > \epsilon) \leq C/(N\epsilon^2)$.

Apply this to $N_k = 2^k$. Since $\sum 2^{-k} < \infty$, $\sum_k P(|S_{2^k}/2^k - 1/b| > \epsilon) < \infty$. The first Borel–Cantelli lemma implies $S_{2^k}/2^k \rightarrow 1/b$ almost surely.

For $2^k \leq N < 2^{k+1}$, one has $S_{2^k} \leq S_N \leq S_{2^{k+1}}$. Dividing appropriately and using $S_{2^k}/2^k \rightarrow 1/b$ yields $S_N/N \rightarrow 1/b$ almost surely.

Step 2: Frequency of a fixed block.

Let $w = (a_1, \dots, a_m)$ be a block of length m . Define $J_n = 1\{(X_n, \dots, X_{n+m-1}) = w\}$ and $T_N = \sum_{n \leq N} J_n$. Then $E[J_n] = b^{-m}$. Moreover J_n and J_r are independent whenever $|n-r| \geq m$.

Since only $O(N)$ pairs contribute nonzero covariance, $\text{Var}(T_N) = O(N)$. Therefore Chebyshev gives $P(|T_N/N - b^{-m}| > \epsilon) \leq C/(N\epsilon^2)$. Applying this to $N = 2^k$ and using Borel–Cantelli again yields $T_{2^k}/2^k \rightarrow b^{-m}$ almost surely.

Interpolating between successive powers of two shows $T_N/N \rightarrow b^{-m}$ almost surely.

Step 3: Conclusion.

There are only countably many finite blocks. For each block w , the event $\{T_N(w)/N \rightarrow b^{-|w|}\}$ has probability one. Taking the countable intersection over all blocks preserves probability one. Consequently, for almost every x , every finite block appears with the correct limiting frequency.

Hence the set of normal numbers in base b has Lebesgue measure one: $\lambda(\{x : x \text{ is normal to base } b\}) = 1$.